

Mathematical Primer

For “Prediction Markets Are Fisher Markets”

Five ideas + the full proof of Theorem 5

What this document covers. The paper uses five mathematical ideas. Each builds on the previous. §6 then walks through the full Theorem 5 proof — the paper’s main result — applying all five ideas.

#	Idea	Unlocks
1	Complementary slackness	Theorems 3, 4, 5
2	Fenchel conjugates	§2.2, §2.3, §2.8 (Theorems 1, 2, 4)
3	KKT conditions	§2.4, §5.2 (Theorems 3, 5)
4	The budget absorption trick	Theorem 5 part (3) — the punchline
5	Why log makes it convex	The entire §3 → §5 arc
6	Theorem 5 step by step	The full proof with cash variable

1. Minting Cost and the LP: Why $\max_k D_k$?

Before the five ideas, one piece of setup from the paper (§2.1) needs unpacking: the clearing LP and its minting cost.

1.1. What Minting Is

A prediction market has K mutually exclusive outcomes. Minting creates one share of *every* outcome at cost \$1. This is fairly priced: exactly one outcome resolves to \$1 at settlement, so a complete set is always worth \$1.

The key constraint: you cannot create shares of a single outcome. Every mint produces one share of each.

1.2. What D_k Actually Is

D_k looks like “demand for outcome k ” but it’s more precise than that. It is the *net* creation requirement — how many new shares of outcome k the exchange must create, after netting buys against sells:

$$D_k = \sum_{i \in \text{buy}(k)} q_i - \sum_{j \in \text{sell}(k)} q_j$$

Some shares come from sellers (they already hold shares and are giving them up). D_k is what’s left over — the shortfall that sellers can’t cover.

Example. Outcome A has 100 shares of buy orders filled and 40 shares of sell orders filled. Net: $D_A = 100 - 40 = 60$. The exchange must *create* 60 new A-shares. The other 40 came from sellers.

Crucially, **the optimizer controls D_k** through its choice of fills $\mathbf{q}$. It can make D_k small by filling fewer buy orders or filling more sell orders. D_k is not external demand — it’s a consequence of the optimizer’s decisions.

If $D_k \leq 0$ (more selling than buying), no new shares of outcome k are needed — shares are being returned/destroyed.

1.3. Why $M \geq D_k$

M is the number of complete sets minted. Each mint produces one share of *every* outcome. After M mints, you have M new shares of each outcome available.

D_k is how many new shares of outcome k the fills require (as computed above).

The constraint $M \geq D_k$ says: **new shares produced $\geq$ new shares needed**, for each outcome. You must mint enough complete sets to cover the largest shortfall. You don't need $M \geq D_k$ for outcomes where sellers already cover the demand — only where the net is positive.

1.4. The LP Is a Tradeoff

The clearing LP is:

$$\max_{\mathbf{q}, M} \quad \underbrace{\sum_i w_i q_i}_{\text{welfare from fills}} - \underbrace{M}_{\text{minting cost}} \quad \text{s.t.} \quad D_k(\mathbf{q}) \leq M \quad \forall k, \quad \mathbf{q} \in [0, \overline{Q}]$$

The optimizer chooses *both* which orders to fill *and* how much to mint. It's a cost–benefit tradeoff:

- **Filling orders creates welfare** ($w_i q_i > 0$ for profitable orders).
- **Filling orders creates net demand** (D_k increases), which requires minting.
- **Minting costs money** ($-M$ in the objective).
- **The optimizer is free to fill nothing.** Setting $\mathbf{q} = \mathbf{0}$ gives $D_k = 0$, $M = 0$, objective = 0. This is always feasible.

The optimizer fills orders only when the welfare gained exceeds the minting cost incurred. The whole LP is the optimizer asking: **which fills create enough welfare to justify the minting they require?**

Example. Suppose filling a buy order creates \$0.60 of welfare but requires minting that costs \$0.80 (because no sellers offset it). The optimizer skips this order — it destroys value. But if a sell order offsets half the minting need, the net cost drops to \$0.40, and the fill becomes worthwhile.

1.5. Why $M = \max_k D_k$ at the Optimum

The constraint *allows* minting more than needed ($M \geq D_k$, not $M = D_k$). But M is subtracted from the objective — it's a cost. The optimizer pushes M down as far as possible, which means $M = \max_k D_k$ at the optimum (the tightest the constraints allow). The $\geq$ is the LP formulation; the optimizer tightens it to equality.

This is also complementary slackness at work: the dual variable of $D_k \leq M$ is the price p_k , and $p_k > 0$ only where $D_k = M$ (the binding constraints).

1.6. Why Is the Surplus Free?

When you mint 70 sets but only need 30 B-shares, the remaining 40 B-shares are surplus. Why don't they cost anything? Because complete sets cost \$1 and pay out \$1 — the surplus shares of losing outcomes are worthless at settlement, and the one winning outcome's surplus was already paid for by the mint. The $\max_k D_k$ accounting captures this exactly: you pay for the most-demanded outcome, and everything else comes free with the complete sets.

2. Complementary Slackness

You have a constraint — say, a fill can’t exceed its maximum quantity:

$$q_i \leq \bar{Q}_i$$

The optimizer assigns a *dual variable* (also called a multiplier) μ_i^+ to this constraint. Think of it as the “price” of the constraint: how much better would the objective be if you relaxed this bound by one unit?

Complementary slackness says:

Rule. At the optimum, at least one of these is zero: the multiplier, or the slack.

$$\mu_i^+ \cdot (\bar{Q}_i - q_i) = 0$$

Two cases:

Situation	Slack	Multiplier
Constraint is tight ($q_i = \bar{Q}_i$)	Zero — no room left	Can be positive — relaxing would help
Constraint is loose ($q_i < \bar{Q}_i$)	Positive — room left	Must be zero — relaxing is pointless

Intuition. “Relaxing” means raising the ceiling — changing $\bar{Q}_i$ from 100 to 101, giving the constraint more room. If you’re not using the full capacity, a higher ceiling wouldn’t help — so the “price” of more capacity is zero. If you *are* at full capacity, a higher ceiling might actually help, so the price can be positive.

Where the paper uses this:

- **Theorem 3** (clearing prices): the dual variable of the minting constraint $D_k \leq M$ is the price p_k . Complementary slackness says $p_k > 0$ only where $D_k = M$ — price is positive only for the highest-demand outcome.
- **§2.4** (clearing rule): gives the three cases (fully filled / unfilled / marginal).
- **Theorem 5** (budget absorption): the $\lambda_i^+ q_i$ terms that make spending $\leq$ budget.

3. Fenchel Conjugates

This is the engine of §2. A Fenchel conjugate transforms a function from “quantity space” to “price space.”

3.1. The Definition

Given a function $f(x)$, its conjugate is:

$$f^*(p) = \sup_x [p \cdot x - f(x)]$$

You sweep over all x , computing “revenue minus cost,” and take the best.

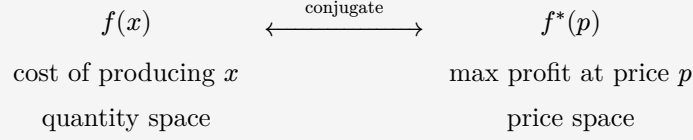
3.2. The Meaning

Think of $f(x)$ as a *cost function* — producing quantity x costs $f(x)$. Now someone offers price p per unit. Your profit from producing x is:

$$\text{profit} = \underbrace{p \cdot x}_{\text{revenue}} - \underbrace{f(x)}_{\text{cost}}$$

The conjugate $f^*(p)$ is your **maximum profit at price p** — the best you can do by choosing the optimal production quantity.

Key idea. f lives in quantity space. f^* lives in price space. They encode the same information from dual perspectives. For convex f , the conjugate of the conjugate gives back f .



3.3. Worked Example: Minting Cost (Theorem 1)

Our cost function is $V(\mathbf{D}) = \max_k D_k$ — the minting cost. We want the conjugate:

$$V^*(\mathbf{p}) = \sup_{\mathbf{D}} \left[\sum_k p_k D_k - \max_k D_k \right]$$

You're selling outcome shares at prices p_k and paying minting cost $\max_k D_k$. What demand vector $\mathbf{D}$ maximizes profit?

Case: $\mathbf{p}$ is a probability vector ($\sum p_k = 1$, all $p_k \geq 0$).

Revenue = $\sum p_k D_k$ is a weighted average of the D_k 's with weights summing to 1. A weighted average can never exceed the maximum:

$$\sum p_k D_k \leq \max_k D_k$$

So profit ≤ 0 for every $\mathbf{D}$. At $\mathbf{D} = \mathbf{0}$, profit = 0. Best profit: $V^*(\mathbf{p}) = 0$.

Case: $\sum p_k > 1$.

Set all $D_k = t$, send $t \rightarrow \infty$. Revenue = $t \sum p_k$. Cost = t . Profit = $t(\sum p_k - 1) \rightarrow \infty$.

Arbitrage: you mint complete sets at \$1 each and sell the shares for $\sum p_k > \$1$. Scale up for unbounded profit.

Case: $\sum p_k < 1$.

Set all $D_k = -t$ (buy back shares), send $t \rightarrow \infty$. Profit = $t(1 - \sum p_k) \rightarrow \infty$.

Reverse arbitrage: buy a complete set of shares for $\sum p_k < \$1$ and redeem for \$1.

Case: some $p_k < 0$.

Drive $D_k \rightarrow -\infty$ for that outcome. Someone is paying you to take shares. Infinite profit.

Result:

$$V^*(\mathbf{p}) = \begin{cases} 0 & \text{if } \mathbf{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

This is the simplex indicator. Its meaning: **the only prices where you can't arbitrage the minting mechanism are probabilities**. The probability axiom is a no-arbitrage condition, not a modeling choice.

3.4. Worked Example: LMSR Cost (Theorem 2)

Replace minting cost with LMSR: $C_b(\mathbf{D}) = b \ln \sum \exp(D_k/b)$. Same game:

$$C_b^*(\mathbf{p}) = \sup_{\mathbf{D}} \left[\sum p_k D_k - b \ln \sum \exp(D_k/b) \right]$$

For the minting cost (Theorem 1), $V = \max_k D_k$ has a kink — it's not differentiable where two D_k 's are tied — so we had to reason case-by-case. The LMSR cost C_b is smooth (that's the whole point of entropy smoothing), so we can use calculus: find the maximum by setting the derivative to zero.

The expression inside the sup — “linear in $\mathbf{D}$ ” minus “convex in $\mathbf{D}$ ” — is *concave* in $\mathbf{D}$. For a concave function, any point where the derivative is zero is the global maximum (not just a local one — this is what convexity/concavity buys you). So setting the derivative to zero finds *the* answer:

$$\begin{aligned} \frac{\partial}{\partial D_k} \left[\sum p_k D_k - C_b(\mathbf{D}) \right] &= p_k - \frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)} = 0 \\ \therefore p_k &= \frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)} \end{aligned}$$

This is the **softmax**. The conjugate asks “what demand vector maximizes profit at prices $\mathbf{p}$?” and the answer is: the one where prices equal the softmax of demand.

Invert ($D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp(D_j/b)$), substitute back, and after algebra:

$$C_b^*(\mathbf{p}) = b \sum_k p_k \ln p_k$$

Negative Shannon entropy. Finite only on the simplex (softmax always outputs probabilities). Most negative at uniform $p_k = 1/K$ (maximum entropy = most uncertain prices).

3.5. The Duality Picture

Cost (quantity space)		Conjugate (price space)	Character
$V = \max_k D_k$	$\longleftrightarrow$	$V^* = \delta_{\Delta}$	Hard wall: probabilities or nothing
$C_b = b \ln \sum \exp(D_k/b)$	$\longleftrightarrow$	$C_b^* = b \sum p_k \ln p_k$	Soft penalty: non-uniform costs more

As $b \rightarrow 0$: the soft penalty hardens into the wall.

Why this matters for the paper. In §2.8, the clearing problem is rewritten in price space as:

$$\min_{\mathbf{p} \in \Delta} [W^*(\mathbf{p}) + C_b^*(\mathbf{p})]$$

The entropy term C_b^* is *strictly convex* — it curves. That curvature forces a unique minimizer. This is Theorem 4: without budgets, prices are always unique. The entire §3 obstacle is about budgets breaking this.

4. KKT Conditions

You want to maximize $f(x)$ subject to $x \in [0, \bar{Q}]$. Calculus says “set $f'(x) = 0$ ” — but what if the max is at the boundary?

KKT handles boundary optima with multipliers:

$$f'(x) = \mu^+ - \mu^-$$

where $\mu^+, \mu^- \geq 0$ with complementary slackness ($\mu^+(x - \bar{Q}) = 0$, $\mu^-x = 0$).

Case	Condition	Meaning
Interior ($0 < x < \bar{Q}$)	$f'(x) = 0$	Just calculus — both multipliers zero
At ceiling ($x = \bar{Q}$)	$f'(x) \geq 0$	Function still rising — you’d want more but you’re capped
At floor ($x = 0$)	$f'(x) \leq 0$	Function falling — you want none

In the paper (§2.4). The objective per order is $w_i q_i - C_b(\mathbf{D})$. For buy order i on outcome k :

$$\frac{\partial}{\partial q_i} [w_i q_i - C_b] = L_i - \underbrace{\frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)}}_{p_k}$$

The derivative is just $L_i - p_k$ (limit price minus clearing price). KKT says:

Case	Derivative	In English
Fully filled ($q_i = \bar{Q}_i$)	$L_i - p_k \geq 0$	Limit above price — fill everything
Unfilled ($q_i = 0$)	$L_i - p_k \leq 0$	Limit below price — fill nothing
Marginal ($0 < q_i < \bar{Q}_i$)	$L_i - p_k = 0$	Limit equals price — this order sets the price

This is the Uniform Clearing Price rule: buy if your limit exceeds the price.

Why this matters. Theorem 5 uses the exact same KKT structure, but with B_k/U_k replacing w_i . If you understand the above, you understand the Theorem 5 proof — it’s the same three cases with one extra chain-rule step.

5. The Budget Absorption Trick

This is the punchline of the paper (Theorem 5, part 3). We replace linear welfare $\sum w_i q_i$ with $B_k \ln U_k$ where $U_k = \sum_{i \in \text{MM}_k} L_i q_i$ is MM k ’s total weighted fill. An important modeling choice: MM_k contains only MM k ’s *buy* orders (with $L_i > 0$). Sell orders from MMs are treated as retail (linear welfare). This ensures $U_k > 0$ and aligns with the Fisher market analogy, where agents are consumers (buyers of goods).

5.1. Step 1: KKT for the Log Objective

For MM buy order i belonging to MM k , differentiate $B_k \ln U_k$ with respect to q_i :

$$\frac{\partial}{\partial q_i}[B_k \ln U_k] = B_k \cdot \frac{1}{U_k} \cdot L_i$$

This is the chain rule: derivative of $\ln$ is $1/U_k$, derivative of U_k w.r.t. q_i is L_i , times the weight B_k .

The full KKT also includes the minting cost derivative. Since this is a buy order, it increases $D_{m(i)}$ by q_i , costing $p_{m(i)}$ per share. With box multipliers:

$$\frac{B_k L_i}{U_k} - p_{m(i)} = \lambda_i^+ - \lambda_i^-$$

5.2. Step 2: The Telescoping

Multiply both sides by q_i :

$$\frac{B_k L_i q_i}{U_k} - p_{m(i)} q_i = \lambda_i^+ q_i - \lambda_i^- q_i$$

Note: $\lambda_i^- q_i = 0$ by complementary slackness (if $q_i > 0$ then $\lambda_i^- = 0$; if $\lambda_i^- > 0$ then $q_i = 0$). So the last term vanishes. Now sum over all $i \in \text{MM}_k$:

$$\frac{B_k}{U_k} \cdot \underbrace{\sum_{i \in \text{MM}_k} L_i q_i}_{\text{this is } U_k \text{ by definition}} - \underbrace{\sum_{i \in \text{MM}_k} p_{m(i)} q_i}_{\text{capital spent on purchases}} = \underbrace{\sum_{i \in \text{MM}_k} \lambda_i^+ q_i}_{\geq 0}$$

The left side: U_k cancels, leaving $B_k - \sum p_{m(i)} q_i$.

$$\begin{aligned} B_k - \sum_{i \in \text{MM}_k} p_{m(i)} q_i &= \sum_{i \in \text{MM}_k} \lambda_i^+ q_i \geq 0 \\ \therefore \sum_{i \in \text{MM}_k} p_{m(i)} q_i &\leq B_k \end{aligned}$$

Each MM's capital deployed on purchases is **at most** B_k , with equality when no fill hits its upper bound.

That's it. Five lines of algebra. The budget constraint was never imposed — it *emerged* from the first-order conditions of the log objective.

5.3. Why This Works: Intuition

$\ln(U_k)$ has a singularity at $U_k = 0$: the slope goes to ∞ .

Near $U_k = 0$: Marginal value B_k/U_k is huge — fill desperately

As U_k grows: Marginal value B_k/U_k shrinks — diminishing returns

At optimum: Marginal value per dollar = price for every active order

Because of the B_k scaling, this balance point is reached exactly when total spending = B_k . Bigger budget $\rightarrow$ more fill before saturation. The budget isn't a constraint; it's a *consequence of the curvature*.

6. Why Log Makes It Convex

Now we connect everything. The risk-neutral model has a non-convexity problem (§3). The log model doesn't (§5). Here's why.

6.1. The Risk-Neutral Problem

Objective: $\sum w_i q_i - C_b(D)$.

Component	Curvature	Source
$\sum w_i q_i$ (linear welfare)	None — flat	No help
$-C_b$ (minting cost)	Concave, strength $O(1/b)$	Entropy smoothing
Budget constraint $c(p) \cdot q \leq B_k$	Non-convex, strength $O(1/b)$	Price $\times$ quantity

The only curvature comes from $-C_b$, which scales as $O(1/b)$. But the budget non-convexity *also* scales as $O(1/b)$ — they grow at the same rate. Neither dominates. **Deadlock.**

At $b = 0$ (LP), there is zero curvature and the budget constraint is fully non-convex. No hope.

6.2. The Risk-Averse Fix

Objective: $\sum B_k \ln U_k + \text{retail} - C_b(D)$.

Component	Curvature	Source
$\sum B_k \ln U_k$ (log welfare)	Strictly concave, $O(B_k/U_k^2)$	$\ln$ — independent of b
$-C_b$ (minting cost)	Concave, strength $O(1/b)$	Entropy smoothing — bonus
Budget constraint	Gone	Absorbed by §4 trick

Two things change simultaneously:

1. **The budget constraint disappears** from the feasible set. The bilinear $c(p) \cdot q \leq B_k$ is gone — absorbed into the objective (§4 trick). No more non-convex feasible set.
2. **Strict curvature appears** in the objective. The $\ln$ provides $O(B_k/U_k^2)$ concavity, independent of b . This guarantees a unique maximum.

The $-C_b$ term is now pure bonus — extra concavity on top of what $\ln$ already provides. Even at $b = 0$ (no entropy smoothing at all), the program is still strictly concave:

$$\max_{q \in \mathcal{C}} \sum_k B_k \ln U_k + \sum_{j \notin \text{MM}} w_j q_j - \max_k D_k$$

The $-\max_k D_k$ is concave (not strictly), but $\ln$ carries the strict concavity alone. **No annealing needed.**

6.3. The Full Picture

	Risk-neutral	Risk-averse
Welfare	$\sum w_i q_i$ (linear)	$\sum B_k \ln U_k$ (log)
Budget constraint	Explicit, bilinear, non-convex	Absent — absorbed into objective
Curvature source	$-C_b$ only ($O(1/b)$)	$\ln$ ($O(B_k/U_k^2)$, b -independent)

At $b = 0$	Zero curvature, non-convex — hopeless	Strict curvature, convex — fine
Unique prices?	Not always (Prop. 2)	Always (Thm. 5)
Complexity	GNEP (intractable in general)	Convex program (polynomial)

One modeling change — linear welfare $\rightarrow$ log welfare — removes the non-convexity and adds strict curvature. The paper’s thesis: this isn’t a trick. The linear model was the fiction.

7. Theorem 5 Step by Step

Sections 1–5 gave you the five tools. This section applies all of them to walk through the actual proof of Theorem 5 — the paper’s main result. Everything below is a line-by-line unpacking of §5.2 in the paper.

7.1. The Program

The risk-averse clearing program is:

$$P_b^{\text{RA}} : \max_{q \in \mathcal{C}, s \geq 0} \underbrace{\sum_k [B_k \ln(U_{k(q)} + s_k) - s_k]}_{\text{MM welfare}} + \underbrace{\sum_{j \notin \text{MM}} w_j q_j}_{\text{retail welfare}} - \underbrace{C_{b(D(q))}}_{\text{minting cost}}$$

where $U_k = \sum_{i \in \text{MM}_k^+} L_i q_i$ is MM k ’s total weighted fill (buy orders only, $L_i > 0$) and $s_k \geq 0$ is retained cash.

Three things to unpack about the per-MM objective $B_k \ln(U_k + s_k) - s_k$:

Why $U_k + s_k$, not just U_k ? If you used $B_k \ln U_k$ alone (the simplified version from §4), an over-capitalized MM ($B_k > U_k^{\text{LP}}$) would be *forced* to spend all its budget on fills to maximize $\ln U_k$ — including unprofitable ones. The cash variable s_k is a release valve: the MM can park unused budget as cash instead of buying garbage fills.

Why $-s_k$? Cash has linear disutility — each dollar retained costs exactly one dollar of welfare. This makes cash a *numeraire good*: it has constant marginal value 1, so the MM only retains cash when no fill can beat that marginal value. Without $-s_k$, the MM would hoard infinite cash for free.

Why B_k out front? This scales the curvature. Bigger budget $\rightarrow$ the $\ln$ singularity “fires” at a higher U_k . As §4 showed, the B_k scaling is exactly what makes total spending equal B_k at the optimum — the budget is encoded in the curvature, not in a constraint.

7.2. Part 1: Strict Concavity (Why a Unique Optimum Exists)

Claim: The objective is strictly concave in (q, s) , so it has a unique maximizer.

The composition rule. There is a standard fact from convex analysis:

If g is strictly concave and increasing, and h is affine (linear + constant), then $g \circ h$ is strictly concave.

Why? Take two points x_1, x_2 and $\lambda \in (0, 1)$. Affinity of h gives $h(\lambda x_1 + (1 - \lambda)x_2) = \lambda h(x_1) + (1 - \lambda)h(x_2)$. Then strict concavity of g gives $g(\lambda h(x_1) + (1 - \lambda)h(x_2)) > \lambda g(h(x_1)) + (1 - \lambda)g(h(x_2))$. That's the definition of strict concavity for $g \circ h$.

Apply this to our objective:

Piece	Role	Curvature
$U_k + s_k = \sum L_i q_i + s_k$	h — affine in $(\mathbf{q}, s_k)$	None (flat)
$\ln(\cdot)$	g — strictly concave, increasing	Strict
$B_k \ln(U_k + s_k)$	$B_k \cdot g \circ h$	Strictly concave ($B_k > 0$)
$-s_k$	Linear	None (preserves concavity)
Retail welfare	Linear in $\mathbf{q}$	None (preserves concavity)
$-C_b$	Concave (not strictly)	Bonus curvature

Sum of a strictly concave function and concave functions is strictly concave. The $\ln$ alone carries the strict part.

The maximum is attained (not just a supremum). Two boundary arguments:

- As $s_k \rightarrow \infty$: $B_k \ln(s_k) - s_k \rightarrow -\infty$ because $-s_k$ grows faster than $\ln s_k$. So the objective is *coercive* in s_k — it eventually drives the objective down.
- As $U_k + s_k \rightarrow 0^+$: The gradient $\frac{\partial}{\partial s_k}[B_k \ln(U_k + s_k) - s_k] = B_k/(U_k + s_k) - 1 \rightarrow +\infty$. The slope becomes infinitely steep upward. So the optimizer is always pushed *away* from zero — the singularity repels.

These two forces (coercive at infinity, repulsive at zero) trap the optimum in a bounded region. Bounded region + continuous function $\rightarrow$ maximum attained. Strict concavity $\rightarrow$ the maximizer is unique.

7.3. Part 2: Limit Order Exactness ($\mu_k \leq 1$)

Define the **shadow price of capital**:

$$\mu_k \equiv \frac{B_k}{U_k + s_k}$$

This is the marginal value of one more dollar of fills-plus-cash to MM k . Now differentiate the objective w.r.t. the cash variable s_k :

$$\frac{\partial}{\partial s_k}[B_k \ln(U_k + s_k) - s_k] = \underbrace{\frac{B_k}{U_k + s_k}}_{\mu_k} - 1 = \mu_k - 1$$

The variable s_k has the constraint $s_k \geq 0$ (you can't retain negative cash). Apply KKT (§3 of this primer) to this floor constraint:

Case	Condition	Meaning
Interior ($s_k > 0$)	$\mu_k - 1 = 0$, so $\mu_k = 1$	Marginal fill = marginal cash = 1
At floor ($s_k = 0$)	$\mu_k - 1 \leq 0$, so $\mu_k \leq 1$	Fills are worth less than cash — hold none

In both cases: $\mu_k \leq 1$. This is unconditional — it holds for every MM at every optimum.

Why this matters. The KKT condition for fill q_i of MM k (with minting cost included) is:

$$\mu_k L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$$

For a fill to happen ($q_i > 0$), the left side must be ≥ 0 (by KKT at the floor, same logic as §3). So $\mu_k L_i \geq p_{m(i)}$, which gives:

$$L_i \geq \frac{p_{m(i)}}{\mu_k} \geq p_{m(i)}$$

The limit price must exceed the clearing price. **No order fills below its stated price.** The $\mu_k \leq 1$ guarantee is what makes this work — it's the cash variable doing its job.

7.4. The Two Regimes

Complementary slackness on $s_k \geq 0$ creates two clean regimes:

Regime	s_k	μ_k	Behavior
Capital-constrained ($U_k > B_k$)	0	$B_k/U_k < 1$	Throttles low-ROI fills
Over-capitalized ($U_k < B_k$)	$B_k - U_k > 0$	1	Acts like LP

Capital-constrained regime. The MM has more profitable opportunities than budget. The shadow price $\mu_k < 1$ scales down all limit prices by a factor μ_k . An order with $L_i = 0.55$ and $\mu_k = 0.86$ acts as if its limit is $\mu_k L_i = 0.47$. High-ROI orders (high L_i) survive the scaling; low-ROI orders get priced out. This is the MM organically prioritizing its best fills.

Over-capitalized regime. The MM has more budget than opportunities. It sets $s_k = B_k - U_k > 0$ (parks the excess as cash) and $\mu_k = 1$. The fill KKT becomes $L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$ — exactly the risk-neutral LP KKT. The log model *degenerates to the LP* when the budget doesn't bind. This is Proposition 3 (LP Recovery).

7.5. Part 3: Budget Absorption (With Cash)

§4 showed the absorption trick for the simplified $B_k \ln U_k$ model. Here we do it with s_k . The algebra is nearly identical, with one extra step.

Start from the fill KKT (from Part 2):

$$\mu_k L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$$

Multiply both sides by q_i and use $\lambda_i^- q_i = 0$ (complementary slackness — same as §4):

$$\mu_k L_i q_i - p_{m(i)} q_i = \lambda_i^+ q_i$$

Sum over all $i \in \text{MM}_k^+$:

$$\underbrace{\mu_k \sum L_i q_i}_{U_k} - \sum p_{m(i)} q_i = \underbrace{\sum \lambda_i^+ q_i}_{\geq 0}$$

So $\sum p_{m(i)} q_i \leq \mu_k U_k$. (Same as §4 so far.)

The extra step. By definition, $\mu_k = B_k / (U_k + s_k)$, so $\mu_k (U_k + s_k) = B_k$, which gives $\mu_k U_k = B_k - \mu_k s_k$. Since $\mu_k \leq 1$ (Part 2) and $s_k \geq 0$:

$$\mu_k U_k = B_k - \mu_k s_k \leq B_k$$

Combining with $\sum p_{m(i)} q_i \leq \mu_k U_k$:

$$\sum_{i \in \text{MM}_k^+} p_{m(i)} q_i + s_k \leq B_k$$

Total deployment — capital on fills plus retained cash — is at most B_k . The gap $\sum \lambda_i^+ q_i$ is infra-marginal surplus from fully-filled orders — profit that doesn't require additional capital.

The budget constraint was never imposed. It emerged from the KKT conditions of the ln objective plus the cash variable.

7.6. Part 4: Price Uniqueness

For $b > 0$: clearing prices are $p_k = \text{softmax}(\mathbf{D}/b) = \exp(D_k/b) / \sum_j \exp(D_j/b)$. This is a continuous function of $\mathbf{D}$, which is a linear function of $\mathbf{q}$. Since $\mathbf{q}^*$ is unique (Part 1), $\mathbf{D}^*$ is unique, and $\mathbf{p}^*$ is unique.

For $b = 0$: prices are LP duals — elements of the subdifferential of $\max_k D_k$ at $\mathbf{D}^*$. When two or more D_k tie at $\max_j D_j$, the subdifferential is a *face* of the simplex (any convex combination of the tying outcomes' unit vectors works). Fills are still unique, but prices may not be. Any $b > 0$, no matter how small, breaks ties via the exponential and restores uniqueness.

7.7. Worked Example

Binary market ($K = 2$), $b \rightarrow 0$. An MM ($B = 60$) posts two buy-Yes orders: A at $L_A = 0.70$, qty 100; B at $L_B = 0.55$, qty 100. A retail trader posts buy-No at $L_C = 0.60$, qty 200.

LP (no budgets). All orders fill. Balanced minting: $D_{\text{Yes}} = D_{\text{No}} = 200$, $p_{\text{Yes}} = 0.50$. Welfare = $(0.70 - 0.50)(100) + (0.55 - 0.50)(100) + (0.60 - 0.50)(200) = 45$. But MM capital consumed = $0.50 \times 200 = 100 > B = 60$. Budget violated.

Risk-averse. The MM is capital-constrained: $U^{\text{LP}} = 0.70(100) + 0.55(100) = 125 > B = 60$, so $s = 0$ and $\mu = B/U$.

The optimizer finds $\mu = 60/70 \approx 0.86$ (only order A fills, $U = 70$). Check the fill KKT:

- Order A : $\mu L_A = 0.86 \times 0.70 = 0.60 \geq p_{\text{Yes}} = 0.50$. Fills. ✓
- Order B : $\mu L_B = 0.86 \times 0.55 = 0.47 < p_{\text{Yes}} = 0.50$. Does not fill. ✗

The shadow price $\mu < 1$ throttles the lower-ROI order.

Minting balances at $D_{\text{Yes}} = D_{\text{No}} = 100$. Welfare = 30. Capital consumed = $0.50 \times 100 = 50 \leq B = 60$, with the gap of 10 being infra-marginal surplus from order A (it was willing to pay 0.70 but only paid 0.50, so $0.20 \times 100 = 20$ of welfare was “free” profit, of which 10 is the budget gap).

This primer covers the mathematical ideas in “Prediction Markets Are Fisher Markets.” For the full proofs, formal statements, and economic motivation, see the paper itself.